

Machine Learning

Tutorial 2

Exercise 1:

The following table lists a dataset containing the details of five participants in a heart disease study, and a target feature RISK, which describes their risk of heart disease. Each patient is described in terms of four binary descriptive features:

- EXERCISE, how regularly do they exercise.
- SMOKER, do they smoke.
- OBESE, are they overweight.
- FAMILY, did any of their parents or siblings suffer from heart disease.

ID	EXERCISE	SMOKER	OBESE	FAMILY	RISK
1	daily	false	false	yes	low
2	weekly	true	false	yes	high
3	daily	false	false	no	low
4	rarely	true	true	yes	high
5	rarely	true	true	no	high

(a) As part of the study, researchers have decided to create a predictive model to screen participants based on their risk of heart disease. You have been asked to implement this screening model using a **random forest**. The three tables below list three bootstrap samples that have been generated from the above dataset. Using these bootstrap samples, create the decision trees that will be in the random forest model (use entropy-based information gain as the feature selection criterion).

ID	EXERCISE	FAMILY	RISK
1	daily	yes	low
2	weekly	yes	high
2	weekly	yes	high
5	rarely	no	high
5	rarely	no	high
Bootstrap Sample A			

ID	SMOKER	OBESE	RISK
1	false	false	low
2	true	false	high
2	true	false	high
4	true	true	high
5	true	true	high
Bootstrap Sample B			

ID	OBESE	FAMILY	RISK
1	false	yes	low
1	false	yes	low
2	false	yes	high
4	true	yes	high
5	true	no	high
Bootstrap Sample C			

(b) Assuming the random forest model you have created uses majority voting, what prediction will it return for the following query:

EXERCISE=rarely, SMOKER=false, OBESE=true, FAMILY=yes

Exercise 2:

This table lists a dataset of the scores students achieved on an exam described in terms of whether the student studied for the exam (STUDIED) and the energy level of the lecturer when grading the student's exam (ENERGY).

ID	STUDIED	ENERGY	SCORE
1	yes	tired	65
2	no	alert	20
3	yes	alert	90
4	yes	tired	70
5	no	tired	40
6	yes	alert	85
7	no	tired	35

Which of the two descriptive features should we use as the testing criterion at the root node of a decision tree to predict students' scores?

Exercise 3:

The following table shows the target feature, OUTCOME, for a set of instances in a small dataset. An ensemble model is being trained using this dataset using boosting. The table also shows the instance distribution weights, w_4 , for this dataset used at the fifth iteration of the boosting process. The last column of the table shows the predictions made by the model trained at the fifth iteration of boosting, M_4 .

ID	OUTCOME	w_4	M_4
1	<i>Bad</i>	0.167	<i>Bad</i>
2	<i>Good</i>	0.047	<i>Good</i>
3	<i>Bad</i>	0.167	<i>Bad</i>
4	<i>Good</i>	0.071	<i>Bad</i>
5	<i>Good</i>	0.047	<i>Good</i>
6	<i>Bad</i>	0.047	<i>Bad</i>
7	<i>Bad</i>	0.047	<i>Bad</i>
8	<i>Good</i>	0.047	<i>Good</i>
9	<i>Bad</i>	0.167	<i>Bad</i>
10	<i>Good</i>	0.071	<i>Bad</i>
11	<i>Bad</i>	0.047	<i>Bad</i>
12	<i>Good</i>	0.071	<i>Bad</i>

- Calculate the error, ϵ , associated with the set of predictions made by the model M_4 given in the table above.
- Calculate the **confidence factor**, α , associated with M_4 .
- Calculate the updated instance distribution, $w[5]$, based on the predictions made by M_4 .

Exercise 4:

The following table shows a set of predictions made by six models in an ensemble and the ground truth of the target feature in a small test dataset, PROGNOSIS.

ID	PROGNOSIS	M_0	M_1	M_2	M_3	M_4	M_5
1	<i>Bad</i>	<i>Bad</i>	<i>Bad</i>	<i>Good</i>	<i>Bad</i>	<i>Bad</i>	<i>Good</i>
2	<i>Good</i>	<i>Good</i>	<i>Good</i>	<i>Good</i>	<i>Bad</i>	<i>Good</i>	<i>Bad</i>
3	<i>Good</i>	<i>Bad</i>	<i>Good</i>	<i>Bad</i>	<i>Good</i>	<i>Good</i>	<i>Good</i>
4	<i>Bad</i>	<i>Bad</i>	<i>Bad</i>	<i>Bad</i>	<i>Bad</i>	<i>Bad</i>	<i>Good</i>
5	<i>Bad</i>	<i>Good</i>	<i>Bad</i>	<i>Good</i>	<i>Bad</i>	<i>Good</i>	<i>Good</i>

- Assuming that these models are part of an ensemble training using **bagging**, calculate the overall output of the ensemble for each instance in the test dataset.
- Measure the performance of this bagged ensemble using **misclassification rate** (**misclassification rate** is discussed in detail in Section 9.3; it is simply the percentage of instances in the test dataset that a model has incorrectly classified).
- Calculate the overall output of the ensemble for each instance in the test dataset. Assuming that these models are part of an ensemble trained using **boosting** and that the confidence factors, α , for the models are as follows:

M_0	M_1	M_2	M_3	M_4	M_5
0.114	0.982	0.653	0.912	0.883	0.233

- Measure the performance of this boosted ensemble using **misclassification rate**.

Exercise 5

The following **Table 0** lists a sample of dataset that contains four descriptive features < **AGE**: a continuous feature listing the age of the individual; **EDUCATION (EDU)**; **EXPERIENCE (EXP)**, **OCCUPATION (OCC)** >; and one target **ANNUAL INCOME** with 3 levels (*low*, *middle*, *high*).

As part of a study, researchers have decided to create a predictive model for the annual income using model ensembles. The **tables 1, 2 and 3** list three bootstrap samples that have been generated from the dataset

ID	AGE	EDU	EXP	OCC	INCOME
1	30	Bachelors	No	Driver	middle
2	50	Bachelors	Yes	Engineer	middle
3	18	High school	No	Farmer	low
4	27	Bachelors	Yes	Engineer	middle
5	34	High school	Yes	Farmer	middle
6	23	High school	No	Officer	low
7	47	High school	Yes	Driver	middle
8	55	High school	Yes	Engineer	middle
9	38	Doctorate	Yes	Engineer	high
10	42	Doctorate	Yes	Engineer	high

Table 0: Annual Income Prediction Dataset

ID	AGE	EDU	INCOME
5	34	High school	middle
2	50	Bachelors	middle
6	23	High school	low
4	27	Bachelors	middle
5	34	High school	middle
6	23	High school	low
7	47	High school	middle
2	50	Bachelors	middle
9	38	Doctorate	high
10	42	Doctorate	high

Table 1: Bootstrap sample 1

ID	EDU	OCC	INCOME
1	Bachelors	Driver	middle
1	Bachelors	Driver	middle
3	High school	Farmer	low
4	Bachelors	Engineer	middle
3	High school	Farmer	low
6	High school	Officer	low
8	High school	Engineer	middle
8	High school	Engineer	middle
10	Doctorate	Engineer	high
10	Doctorate	Engineer	high

Table 2: Bootstrap sample 2

ID	EXP	OCC	INCOME
1	No	Driver	middle
5	Yes	Farmer	middle
3	No	Farmer	low
3	No	Farmer	low
5	Yes	Farmer	middle
6	No	Officer	low
7	Yes	Driver	middle
3	No	Farmer	low
9	Yes	Engineer	high
9	Yes	Engineer	high

Table 3: Bootstrap sample 3

Q1: Calculate the Entropy for each Bootstrap sample.

Q2: What would be the possible thresholds and the optimal one to split the continuous **AGE** feature in the 1st bootstrap sample? (In a table, give a detailed answer for each threshold split using the entropy-based information gain).

Q3: Calculate the information gain for each feature in each Bootstrap sample knowing that each feature has the same remainder in all datasets where it is present. The remainder values are given in **Table 4**. Complete the table.

Feature	Remainder	Information Gain		
		Bootstrap 1	Bootstrap 2	Bootstrap 3
EDU	0.49			
OCC	0.49			
EXP	0.85			

Table 4: Information Gain for each Feature in each Bootstrap Sample

Q4: Create the decision trees that will be in the **Random Forest model** based on the previous calculations.

ID	INCOME	W0	M0	W1	M1	W2	M2
1	middle		middle		middle		middle
3	low		low		middle		low
5	middle		low		middle		low
6	low		low		low		low
10	high		middle		middle		high

Table 5: Weights and Model Predictions for Boosting

	M0	M1	M2
Error (ϵ)			
Confidence (α)			

Table 6: Errors and Confidence Factors

Q5: Table 5 shows the INCOME for a set of selected instances, the instance distribution weights, w_0 , w_1 and w_2 , AND the predictions made by the models M0, M1, M2 trained using Boosting (AdaBoost), Complete Tables 5 and 6 by calculating the weights, w_0 (initial), w_1 and w_2 at the first 3 iterations of the boosting process and the error, ϵ , and the confidence factor, α , associated with each Model M0, M1, M2.